

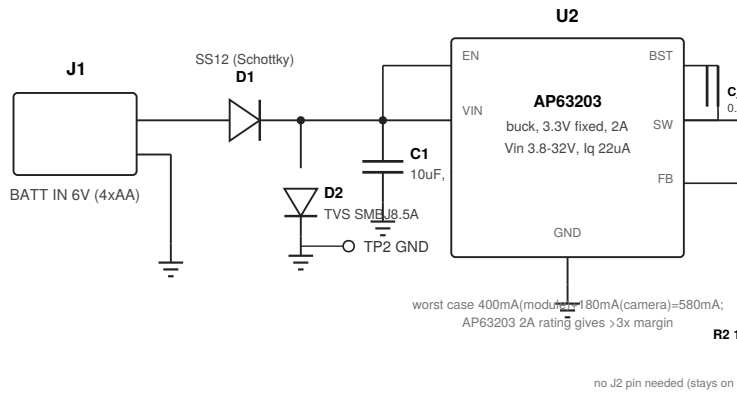
# ESP32-C6-MINI-1-N8 Doorlock + Camera Baseboard — V3.2

50 x 25mm, 2.00mm pitch connectors, programming via J2 (no pogo) • manufactured by Gang Shang • Sheet 1 of 2 (electrical)

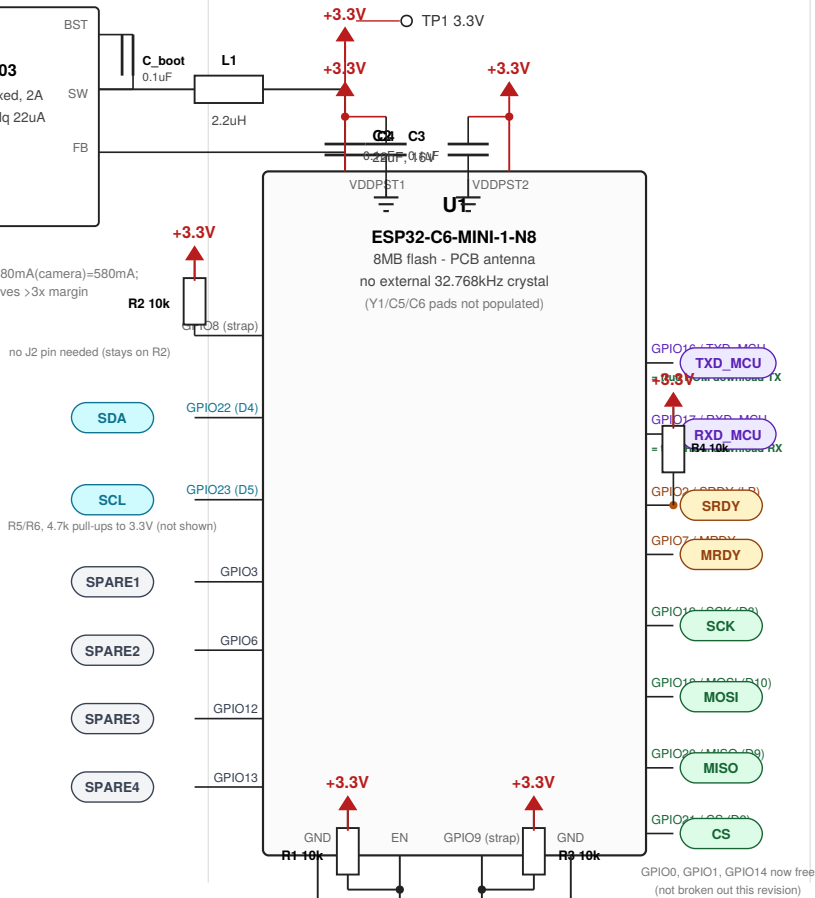
**CRITICAL FIX (still in effect): TXD\_MCU/RXD\_MCU on GPIO16/GPIO17 (was GPIO0/GPIO1) — ROM bootloader's download UART is fixed there, can't be remapped pre-boot (spec §2.1).**

J5 confirmed 6-pin, no RESET line — doorlock MCU owns camera reset via its own UART link, per architecture clarification. GPIO14 is free.

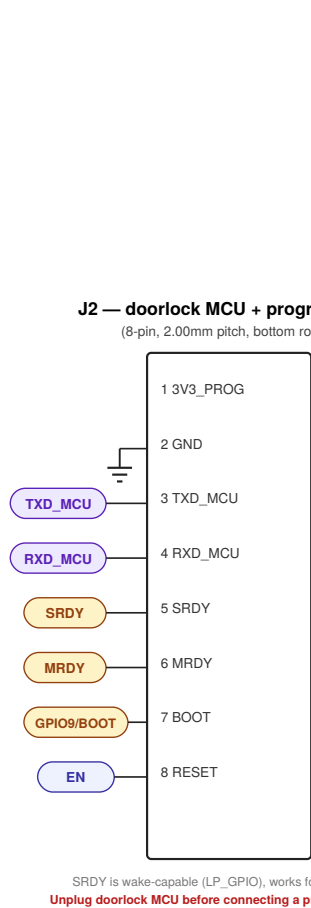
## POWER SUPPLY (buck)



## CORE MODULE



## J2 — MCU + PROGRAMMING (3p)



## NOTES

### Design basis

Modified Seeed XIAO ESP32-C6 R1.0  
Board size 50 x 25mm, mfr: Gang Shang

### 2.1 Programming moved to J2

ROM bootloader's download UART is fixed to GPIO16/17 and can't be remapped pre-boot. TXD\_MCU/RXD\_MCU moved there (was GPIO0/1) so J2 works for both roles. **Unplug MCU before programming.**

### 2.2 Camera RESET intentionally absent

Doorlock MCU is system master and owns camera reset via its own UART link to the camera MCU. This module only relays SPI frame data to Wi-Fi. J5 is 6-pin, GPIO14 free.

### 2.3 SRDY / boot logic unchanged

SRDY still on GPIO2 (deep-sleep wake-capable), BOOT still forces GPIO9 low (not GPIO8) per the documented boot-mode truth table — now via J2 pins 7/8.

### Removed vs. earlier revisions

POGO pads — replaced by J2 pins 1/7/8  
J4 — merged into J3 (I2C + spare GPIO)  
SW1/SW2, Y1/C5/C6, USB-C, charger IC — still omitted, unchanged from V3.1

### Symbol key

ground

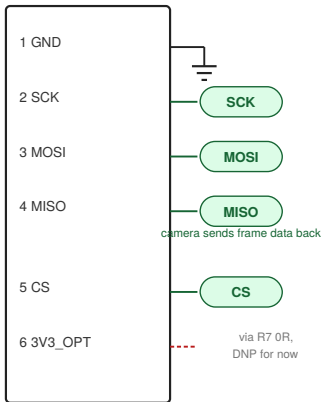
+3.3V  
+3.3V tap  
net label (same color+name = same net)

### Mechanical note

50 x 25mm gives comfortable room: J2+J3 stacked 2-row on the bottom edge, J5 on the left edge, J1 on the right/back. See sheet 2 for the full placement.

## J5 — CAMERA SPI (6p, left edge)

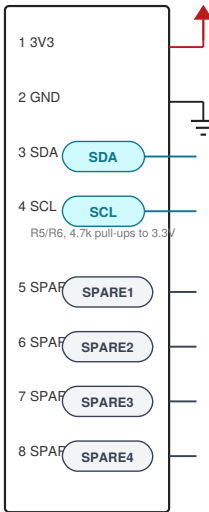
### J5 — camera MCU (SPI, 6-pin)



RESET removed — DL MCU owns camera reset.  
GPIO14 is free (see core module).

## J3 — I2C + SPARE GPIO (8p)

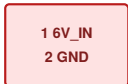
### J3 — I2C + spare GPIO



(merges the old J3+J4, bottom row 2)

## J1 — BATTERY (2p, right/back)

### J1 — battery



2-pin, 2.54mm pitch  
right edge / back side  
placed clear of mounting holes and other connectors

# Mechanical Layout — V3.2 (50 x 25mm, for Gang Shang)

2D top-down placement only — Gang Shang to produce 3D/STEP from this plus connector part numbers • Sheet 2 of 2 (illustrative)



Dashed outlines = illustrative footprints, not exact vendor dimensions. Confirm real connector shroud sizes, antenna keep-out, and mounting-hole diameter directly with Gang Shang's mechanical drawing.

## Placement summary

Bottom edge (25mm height): J2 (row 1, top) + J3 (row 2, bottom), both 8-pin 2.00mm, stacked

Left edge (50mm width): J5, 6-pin 2.00mm

Right edge / back: J1, 2-pin 2.54mm, positioned clear of mounting holes and J2/J5

Top edge: module overhangs by 5mm with a 5mm no-copper keep-out zone above it

**Total connectorized pins: 24 (J1:2 + J2:8 + J3:8 + J5:6) across 4 bodies — comfortable at this board size**

# ESP32-C6-MINI-1-N8 Doorlock + Camera Baseboard — Engineering Specification

V3.2 — 50 x 25mm, 2.00mm-pitch connectors, programming via J2, manufactured by Gang Shang · companion sheets: schematic (p.3) + mechanical layout (p.4)

**Critical fix in this revision:** TXD\_MCU/RXD\_MCU moved to GPIO16/GPIO17 (were GPIO0/GPIO1). The ESP32-C6 ROM bootloader's download UART is fixed to GPIO16/17 and cannot be remapped before firmware runs — programming via J2 on GPIO0/GPIO1 as originally specified would not have worked at all. This is resolved below and reflected in the schematic. Also resolved: J5 confirmed 6-pin, no camera RESET line (doorlock MCU owns camera reset via its own link) — GPIO14 is free.

## 1. Overview

Modified Seeed XIAO ESP32-C6 baseboard, now on a 50 x 25mm outline (grown from the tighter XIAO-compatible footprint of earlier revisions), manufactured by Gang Shang. Connectors are organized as two stacked 8-pin rows on the bottom edge (doorlock MCU/programming, and I2C/spare GPIO), a 6-pin camera SPI header on the left edge, and battery input on the right edge/back. Programming is done directly through J2 — there is no separate pogo-pad or debug connector in this revision.

- **Core:** ESP32-C6-MINI-1-N8, antenna/main-crystal circuitry unchanged from stock XIAO ESP32-C6, 5mm overhang past the top edge
- **Power:** 6V nominal (4xAA) → AP63203 synchronous buck → 3.3V, 2A capability
- **J2 (8-pin, bottom row 1):** doorlock MCU UART1 + SRDY/MRDY handshake + BOOT/RESET, shared with factory/depot programming
- **J3 (8-pin, bottom row 2):** I2C (SDA/SCL) + 4 spare GPIO, merges the prior separate I2C and spare-GPIO connectors
- **J5 (6-pin, left edge):** camera SPI — no reset line; doorlock MCU is system master and owns camera reset independently
- **J1 (2-pin, right edge/back):** battery input, positioned clear of mounting holes and other connectors

## 2. Design notes

### 2.1 Programming moved to J2 — and why it had to move to GPIO16/17

Per Espressif's esptool and ESP-IDF documentation, the ROM bootloader's download UART is fixed to GPIO16 (TX) / GPIO17 (RX) — the chip's default UART0 IO\_MUX pins — and this cannot be reconfigured before any application firmware has run. A specification that routed factory programming through J2 on GPIO0/GPIO1 would have resulted in every unit failing to connect during flashing: the jig would correctly force download mode (GPIO9 low, EN pulsed), but esptool's communication would go out on the wrong pins.

Fix: TXD\_MCU/RXD\_MCU moved to GPIO16/GPIO17. This works for both roles — GPIO16/17 serve as an ordinary UART for doorlock MCU traffic at runtime, and as the ROM bootloader's download UART during programming. **Operational consequence: the doorlock MCU cable must be unplugged from J2 before connecting a programmer** — there's no way to distinguish the two on the wire once they share a pin. This should be documented in assembly/depot procedures.

### 2.2 Camera RESET intentionally omitted

Per architecture clarification: the doorlock MCU is the system's central master and connects directly to the camera MCU via its own UART link. If the camera needs a hardware reset, the doorlock MCU issues it (via its own GPIO or a UART command) — this ESP32-C6 module's role is limited to receiving SPI video frames from the camera and relaying them to Tuya Cloud over Wi-Fi. J5 is 6-pin (GND, SCK, MOSI, MISO, CS, 3V3\_OPT); GPIO14 is free as a result. A 7th-pin option (second GND for EMI shielding on a longer ribbon run) was considered and set aside — revisit only if bring-up shows SPI noise issues or cable sourcing favors it.

### 2.3 SRDY / boot logic unchanged from V3.1

SRDY remains on GPIO2 (the LP\_GPIO/RTC-domain pin confirmed necessary for true deep-sleep wake). BOOT still forces GPIO9 low (not GPIO8) per the documented boot-mode truth table. Both now reached via J2 pins 5 and 7 respectively, instead of pogo pads.

### 3. Mechanical layout

See sheet 2 for the 2D top-down placement. This is an illustrative placement drawing only — Gang Shang's mechanical team should produce 3D/STEP models from this plus the actual connector part numbers chosen.

| Edge                      | Contents                                                         | Fit check                                            |
|---------------------------|------------------------------------------------------------------|------------------------------------------------------|
| Bottom edge (25mm height) | J2 (row 1, top) + J3 (row 2, bottom), both 8-pin 2.00mm, stacked | 8x2.00mm=16mm+~4mm housing=~20mm, fits inside 25mm   |
| Left edge (50mm width)    | J5, 6-pin 2.00mm                                                 | 6x2.00mm=12mm+~4mm housing=~16mm, fits the 50mm edge |
| Right edge / back         | J1, 2-pin 2.54mm                                                 | Placed clear of mounting holes and other connectors  |
| Top edge                  | Module overhangs by 5mm, 5mm no-copper keep-out zone above it    | Confirm exact keep-out against module datasheet      |

4x corner mounting holes, diameter TBD (2.5mm or 3mm assumed — confirm with Gang Shang's mechanical drawing). Total connectorized pins: 24 (J1:2 + J2:8 + J3:8 + J5:6) across 4 bodies — comfortable at this board size, a significant improvement over the tight single-outline layouts in earlier revisions.

### 4. Pin mapping

| U1 pin       | Physical pin | Function                     | Connects to             | Notes                           |
|--------------|--------------|------------------------------|-------------------------|---------------------------------|
| EN (CHIP_PU) | 8            | Reset                        | R1 pull-up 10k; J2 pin8 | unchanged from stock            |
| GPIO8        | 22           | Strap — ROM print control    | R2 pull-up 10k only     | no J2 pin needed                |
| GPIO9        | 23           | Strap — boot mode select     | R3 pull-up 10k; J2 pin7 | must be forced LOW for download |
| GPIO16       | 31           | UART TXD_MCU (+ programming) | J2 pin3                 | true ROM download TX — see §2.1 |
| GPIO17       | 30           | UART RXD_MCU (+ programming) | J2 pin4                 | true ROM download RX — see §2.1 |

| U1 pin               | Physical pin | Function                | Connects to             | Notes                                                        |
|----------------------|--------------|-------------------------|-------------------------|--------------------------------------------------------------|
| GPIO2                | 5            | SRDY, MCU→module wake   | J2 pin5; R4 pull-up 10k | LP_GPIO — deep-sleep wake-capable                            |
| GPIO7                | 16           | MRDY, module→MCU wake   | J2 pin6                 | plain GPIO, no wake requirement                              |
| GPIO18               | 24           | SPI MOSI                | J5 pin3                 | matches stock D10                                            |
| GPIO19               | 25           | SPI SCK                 | J5 pin2                 | matches stock D8                                             |
| GPIO20               | 26           | SPI MISO                | J5 pin4                 | kept — camera sends frame data back                          |
| GPIO21               | 27           | SPI CS                  | J5 pin5                 | matches stock D3                                             |
| GPIO22               | 28           | I2C SDA                 | J3 pin3                 | R5 pull-up 4.7k to 3.3V                                      |
| GPIO23               | 29           | I2C SCL                 | J3 pin4                 | R6 pull-up 4.7k to 3.3V                                      |
| GPIO3                | 6            | Spare                   | J3 pin5                 | clean, no strap function                                     |
| GPIO6                | 15           | Spare                   | J3 pin6                 | clean, no strap function                                     |
| GPIO12               | 17           | Spare                   | J3 pin7                 | clean, no strap function                                     |
| GPIO13               | 18           | Spare                   | J3 pin8                 | clean, no strap function                                     |
| GPIO0, GPIO1, GPIO14 | 12, 13, 19   | Free, not broken out    | —                       | freed by the GPIO16/17 move and RESET removal                |
| GPIO4, GPIO5, GPIO15 | 9, 10, 20    | Reserve, not broken out | —                       | strapping-adjacent — use with boot-time care if needed later |

Physical pin numbers derived directly from the ESP32-C6-MINI-1 module datasheet pin table.

## 5. Net list

| Net          | Nodes                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------|
| VBAT_IN      | J1.1, D1 anode                                                                                   |
| VIN_PROT     | D1 cathode, D2, C1+, U2.VIN, U2.EN                                                               |
| +3.3V        | U2.VOUT(FB node), C2+, U1.VDDPST1, U1.VDDPST2, C3+, C4+, R1-R6 tops, J3.1, J5.6(via R7 DNP), TP1 |
| GND          | J1.2, D2 cathode, C1-, U2.GND, C2-, U1.GND (all), C3-, C4-, J2.2, J3.2, J5.1, TP2                |
| SW node      | U2.SW, L1 pin1, C_boot (to BST)                                                                  |
| EN           | U1.EN, R1 bottom, J2.8                                                                           |
| GPIO8        | U1.GPIO8, R2 bottom                                                                              |
| GPIO9 / BOOT | U1.GPIO9, R3 bottom, J2.7                                                                        |

| Net      | Nodes                                                  |
|----------|--------------------------------------------------------|
| TXD_MCU  | U1.GPIO16, J2.3 (also the factory-programming TX line) |
| RXD_MCU  | U1.GPIO17, J2.4 (also the factory-programming RX line) |
| SRDY     | U1.GPIO2, J2.5, R4 bottom                              |
| MRDY     | U1.GPIO7, J2.6                                         |
| SCK      | U1.GPIO19, J5.2                                        |
| MOSI     | U1.GPIO18, J5.3                                        |
| MISO     | U1.GPIO20, J5.4                                        |
| CS       | U1.GPIO21, J5.5                                        |
| 3V3_OPT  | J5.6, R7 (DNP) to +3.3V                                |
| SDA      | U1.GPIO22, J3.3, R5 top                                |
| SCL      | U1.GPIO23, J3.4, R6 top                                |
| SPARE1-4 | U1.GPIO3/6/12/13, J3.5-8 respectively                  |

## 6. Bill of materials

| Ref    | Qty | Description                                 | Value / part       | Package                 | Example MPN        | Notes                                      |
|--------|-----|---------------------------------------------|--------------------|-------------------------|--------------------|--------------------------------------------|
| U1     | 1   | Wi-Fi 6 / BLE5 / 802.15.4 module            | ESP32-C6-MINI-1-N8 | SMD module, castellated | ESP32-C6-MINI-1-N8 | Unchanged from stock XIAO                  |
| U2     | 1   | Synchronous buck converter, 3.3V            | AP63203            | SOT23-6                 | AP63203WU-7        | 2A, Vin 3.8-32V, Iq 22uA                   |
| L1     | 1   | Buck inductor                               | 2.2uH              | SMD                     | —                  | Confirm value/current rating vs. datasheet |
| C_boot | 1   | Bootstrap capacitor                         | 0.1uF              | 0402/0603               | —                  | Required between BST and SW                |
| D1     | 1   | Schottky diode, reverse-polarity protection | SS12 (1A, 20V)     | SMA / DO-214AC          | SS12-E3/61T        | Unchanged                                  |
| D2     | 1   | Bidirectional TVS, ESD/surge protection     | SMBJ8.5A           | SMB / DO-214AA          | SMBJ8.5A-13-F      | Sized for 6V (4xAA) input                  |
| C1     | 1   | Bulk input capacitor                        | 10μF X5R 16V       | 0805                    | —                  |                                            |
| C2     | 1   | Bulk output capacitor                       | 22μF X5R 16V       | 0805                    | —                  |                                            |

| Ref    | Qty | Description                           | Value / part            | Package        | Example MPN | Notes                                                        |
|--------|-----|---------------------------------------|-------------------------|----------------|-------------|--------------------------------------------------------------|
| C3, C4 | 2   | Decoupling, near U1 power pins        | 0.1μF X7R 16V           | 0402           | —           | Unchanged                                                    |
| R1     | 1   | EN pull-up                            | 10kΩ, 1%                | 0402           | —           |                                                              |
| R2     | 1   | GPIO8 pull-up                         | 10kΩ, 1%                | 0402           | —           |                                                              |
| R3     | 1   | GPIO9 pull-up                         | 10kΩ, 1%                | 0402           | —           |                                                              |
| R4     | 1   | SRDY pull-up                          | 10kΩ, 1%                | 0402           | —           |                                                              |
| R5, R6 | 2   | I2C SDA/SCL pull-ups                  | 4.7kΩ, 1%               | 0402           | —           | On J3                                                        |
| R7     | 1   | Optional camera 3.3V feed             | 0Ω jumper               | 0402           | —           | DNP (do not populate) for now                                |
| J1     | 1   | Battery connector, 2-pin              | 2.54mm pitch            | THT, 2-pin     | —           | Right edge / back                                            |
| J2     | 1   | Doorlock MCU + programming, 8-pin     | 2.00mm pitch            | THT/SMD header | —           | Bottom row 1. Pin1 (3V3_PROG) NC in normal use               |
| J3     | 1   | I2C + spare GPIO, 8-pin               | 2.00mm pitch            | THT/SMD header | —           | Bottom row 2. Merges the prior separate I2C/spare connectors |
| J5     | 1   | Camera MCU SPI, 6-pin                 | 2.00mm pitch            | THT/SMD header | —           | Left edge. No reset line — see §2.2                          |
| —      | —   | Pogo pad cluster                      | REMOVED                 | —              | —           | Replaced by J2 pins 1/7/8                                    |
| —      | —   | SW1, SW2, Y1/C5/C6, USB-C, charger IC | REMOVED / NOT POPULATED | —              | —           | Unchanged from V3.1                                          |

## 7. Power budget (sanity check)

- Worst case: ESP32-C6 active Wi-Fi TX (~300-450mA class) + camera MCU (150-180mA) ≈ 580mA peak — AP63203's 2A rating gives over 3x margin.
- Buck efficiency (92-95% typical) vs. a linear regulator meaningfully reduces heat/battery drain at 6V→3.3V and these currents.
- Idle/quiescent draw dominated by AP63203's I<sub>q</sub> (22μA typical) plus the chip's own sleep current — worth a bench measurement before publishing a battery-life figure.

## 8. Programming & bring-up procedure

- 1 Factory: connect J2 directly (no separate jig connector). Test fixture supplies 3.3V to pin1, pulls pin7 (BOOT/GPIO9) low, pulses pin8 (RESET/EN), then flashes over pins 3/4 (TXD\_MCU/RXD\_MCU = GPIO16/GPIO17, the true ROM download UART).
- 2 No physical buttons and no separate debug connector exist on this board — depot/RMA reprogramming also goes through J2.
- 3 **Before connecting a programmer to J2 in the field, physically disconnect the doorlock MCU cable** — the two share GPIO16/17 and cannot be distinguished electrically.

- 4 After programming, reconnect the doorlock MCU and verify the UART1 + SRDY/MRDY handshake, specifically confirming SRDY wakes the module from deep sleep.
- 5 Connect J5 to the camera MCU and verify SPI behavior independently before final assembly.

## 9. Open items for next revision

- Confirm mounting-hole diameter (2.5mm vs 3mm) and exact connector shroud dimensions directly against Gang Shang's mechanical drawing.
- Confirm antenna keep-out (5mm assumed) against the module datasheet before finalizing copper pour near U1.
- Confirm exact L1 inductor value and C1/C2 values against the AP63203 datasheet's recommended application circuit.
- Document the "unplug doorlock MCU before programming" procedure formally in assembly/depot instructions — see §2.1.
- Decide whether R7 (camera 3.3V feed) ever gets populated, or stays DNP permanently.